

Trade and Spatial Distribution of Economic Activities: Quantitative Analysis of Chinese Manufacturing

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This paper studies the determinants of the spatial distribution of economic activities in China. Specifically, we focus on the different patterns of sectoral growth across Chinese provinces in the last two decades, followed by a series of dramatic policy reforms, including trade liberalization, infrastructure development, productivity growth, and relaxation of mobility restrictions for workers. To understand how those factors interact with each other to shape the spatial distribution of economic activities across regions and sectors, we develop a spatial general equilibrium model of trade with heterogeneous firms (Melitz, 2003; Chaney, 2008). Our model features multiple sectors and multiple regions. We bring the model to the data for the Chinese economy and simulate the counterfactual equilibria to account for the underlying mechanism of regional development.

First, we show the stylized facts of regional development in China following its accession to the WTO, major infrastructure investments, and the reforms of migration restriction known as *hukou*. These policy reforms significantly reduced both inter- and intra-national trade costs and migration costs. Using the firm-level data, which spans the period from 1998 to 2013, we find that the number of firms in labor-intensive sectors (e.g., electronics) grew at an average annual rate of 12.2% in inland provinces, which exceeds the average rate of 6.8% in coastal provinces. In contrast, less labor-intensive and more trade-dependent sectors (e.g., steel) exhibited the opposite pattern, with an average annual rate of 16.1% in coastal provinces compared with 9.5% in inland provinces. Furthermore, contrary to the predictions of the naïve Melitz model, the firm size distribution suggests that the competition is less tight in coastal provinces despite its higher import exposure. Motivated by the descriptive evidence, we develop the multi-sector, multi-region general equilibrium model with heterogeneous firms that features heterogeneity in sectoral production technologies and tradability (Caliendo and Parro, 2014; Balistreri et al., 2011) as well as workers sorting into regions (Redding, 2016; Tombe and Zhu, 2019) and sectors (Bryan and Morten, 2019; Burstein et al., 2019).

We conducted numerical exercises to understand the model mechanism and to examine the key factors shaping the regional distribution of economic activities. With a two-region and two-sector setup, we examined the impacts of lower domestic trade costs on sectoral growth across regions. We specify the parameter values to replicate two representative regions, the coastal (close to the international market) and the inland, and two sectors, labor-intensive electronics and less labor-intensive and more trade-dependent steel. Our result suggests that the labor-intensive sector grows more in the inland region to benefit from lower wages, while the less labor-intensive sector expands to take advantage of better access to international markets. This relocation of economic activities, which is accompanied by the relocation of workers, is consistent with the stylized facts and suggests that the heterogeneity across sectors and regions plays an important role in shaping regional growth in China.

We quantify the model using Chinese firm-level data and inter-regional input-output tables. By characterizing the equilibrium in relative change relative changes, i.e., hat algebra (Dekle et al., 2008), we give counterfactual shocks and stimulate the economy by conditioning on the baseline equilibria in 2002 and 2007. Specifically, we consider three counterfactual scenarios: first, we assess the effects of China's WTO accession by keeping the international trade cost at the pre-WTO level while allowing other fundamentals to evolve as in the data. Second, we focus on the role of domestic infrastructure development by simulating the economy with the 2007 fundamentals while keeping the domestic trade costs fixed at the 2002 level. Third, we fix the

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mobility friction of labor at the 2002 level and see the impacts of *hukou* reforms on the spatial distribution of economic activity. Throughout the counterfactual exercises, we offer new insights into the role of sectors, geography, and labor mobility in shaping regional economic growth and aggregate welfare across China.

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